

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 417 — Past Questions & Materials
by Renance

Course: CHM 417
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 200, 400

This study pack was compiled and restructured by Renance for CHM 417 students at Federal University of Petroleum Resources, Effurun. It brings together 4 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app
Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 4 — CHM 417 - 2024/2025 - First Semester - REACTION MECHANISM

Level 200 • First Semester • PastQuestion

i AIVEMSELY OF PRETO PUM HPS me

PS, tee

nt PAM EMENT OF CHUEMISTRs PPL RUN

pisT SEMESTER EXAMIEN \ TIONS 19242025 ACADEME SF

CUM 407: MEACTION MECHANISM 0 Cyreding WHILE

fiuestion | i WS other iene: t hene ll

ppp mat

inate jiceh®

(juestion 1130 marks)

The tf sing dala WEES

JAL MD Os

for an experiment in the reaction 2A os promlucts

O18 ee 5 ee ee

st GR in| ie

fine (9008) [ae ee.

Determine the rate Constant for the reaction. (10 marks)

b) The following mechanian was se gm phase reaction of chloroform and chlorine (fast)

Step | ty nannies

Step - Chy * CHChigs

, step 3 Che? CChgy

ee is the overall reaction? (2

(iil) Write? aw ;

4) Given that tor a first order re iction, the Gone

M in 60 minutes Calculat ee

(i) The rate oF rea

(it) lhe hall- life of the reaction

e) Thedecompostvon of hy drogen ieside citiye » by

af S00 K and |. 10×10^{-5} J/mol sat 600 K. ' ind £:

Question 2 20 marks)

a) For the fol wens

tion, when the concentration is 0.0 M (3 warks)

(2 marks) — 9

; nas rate tart (9 45% x ity' Linas ,

(3 marks)

-step mechanism.

es of the theory. (2 marks)

'ments made to the theory. by

Kassel (RRK) (iii) Rice, hie.

ction of acetone, $(\text{CH}_3)_2\text{CO}$. in ϕ ous Sole

acetone to form the carbanion GHs ". The

ϕ the product. A simplified version of the m c|

oe Se AS ay

\$ for acetone and A" is its carbanion. Use the steac

* carbanion and agit the rate equation, forthe

END

Paper 2 of 4 — CHM 417 PQ 2

Level 400 • First Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.

b) The following data were obtained for the concentration of product in a reaction $A \rightarrow P$.

Time/secs	10	20	40	60	80	∞
[P]/mol ⁻¹	0.90	1.13	1.29	1.35	1.39	1.50

- Is this a first or second order reaction? Give reasons for your answer.
- Write the rate equation for the above reaction.

Question 4 (20 marks)

a) Cyclopropane isomerizes into propene when heated to 500°C in the gas phase. At low pressures, the following data were obtained.

P/Torr	84.1	11.0	2.89	0.569	0.120	0.067
$10^4 k_{\text{eff}}/\text{s}^{-1}$	2.98	2.23	1.54	0.857	0.392	0.303

Test the Lindemann theory with these data.

b) Outline the refinements made to the Lindemann Theory of unimolecular reactions by the following (i) Hinshelwood (ii) Rice, Ramsperger and Kassel (RRK) (iii) Rice, Ramsperger, Kassel and Marcus (RRKM)

END

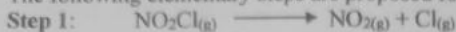
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF CHEMISTRY
FIRST SEMESTER EXAMINATIONS 2022/2023 ACADEMIC SESSION
CHM 417: REACTION MECHANISM (2 Credits)

Instructions: Answer Question 1 and TWO others

Time: 2 hours

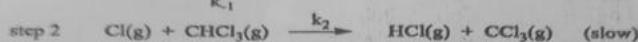
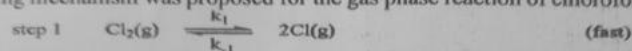
Question 1 (30 marks)

The following elementary steps are proposed for a reaction mechanism:



- Write the overall balanced equation.
- Determine the molecularity of each step.
- Write the rate law for each step
- The rate constant for the decomposition of N_2O_5 is $5.2 \times 10^{-5} \text{ s}^{-1}$ at 25°C and $8.2 \times 10^{-3} \text{ s}^{-1}$ at 75°C . Calculate the activation energy (E_a) of the reaction.
- In the first-order reaction $\text{A} \rightarrow \text{B}$, 25% of A reacts in 42 minutes at 25°C . Calculate the half-life of the reaction.
- Enumerate the essential steps in a chain reaction.

The following mechanism was proposed for the gas phase reaction of chloroform and chlorine.



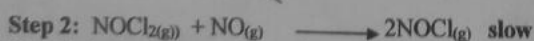
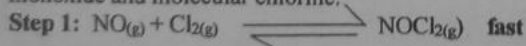
- What are the intermediates in the mechanism?
- What is the rate determining step?
- What is the rate law and overall order of the reaction predicted by this mechanism?
- A proposed mechanism for the reduction of nitrogen as NO by hydrogen is:
Step 1: $\text{H}_{2(g)} + 2\text{NO}_{(g)} \rightarrow \text{N}_2\text{O}_{(g)} + \text{H}_2\text{O}_{(g)}$ (slow)
Step 2: $\text{N}_2\text{O}_{(g)} + \text{H}_{2(g)} \rightarrow \text{N}_2_{(g)} + \text{H}_2\text{O}_{(g)}$ (fast). Write the rate law for the reaction.

Question 2 (20 marks)

- The rate of the reaction $\text{NO}_2 + \text{CO} \rightarrow \text{NO} + \text{CO}_2$ depends only on the concentration of nitrogen dioxide. The following data were collected.

$[\text{NO}_2], (\text{M})$	0.500	0.444	0.381	0.340	0.250	0.174
Time (secs)	0	1200	3000	4500	9000	18000

- What is the order of the reaction? Give reasons.
 - Using graphical method, determine the value of the rate constant for the reaction.
 - Calculate the concentration of NO_2 at 27000 s after the start of the reaction.
- The two-step mechanism below has been proposed for a reaction between nitrogen monoxide and molecular chlorine.



Use this information to derive the equation and predicted rate law for the overall reaction.

Question 3 (20 marks)

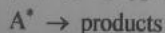
Gaseous, unimolecular decomposition reactions may proceed according to the following mechanism in which the collision between two normal molecules A produces an activated molecule A^* which, in turn, may be deactivated by collision or decompose into products.



rate constant k_1



rate constant k_{-1}



rate constant k_2

- Using the steady state approximation for the quantity $d[\text{A}^*]/dt$, derive the rate law for the reaction $\text{A} \rightarrow \text{products}$.

Paper 3 of 4 — CHM 417 REACTION KINETICS

Level 400 • First Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.

DEPARTMENT OF CHEMISTRY
CHM 417: REACTION KINETICS

Instructions: Answer ALL Questions.

Time: 50 minutes

- a) Derive the rate equation and the half-life expression for the reaction $2A \rightarrow \text{products}$.
b) In the hydrolysis of ethyl acetate using equal concentrations of ester and NaOH, the following results were obtained:

T(minutes)	0	5	15	25	35
HCl (ml)	16.0	10.24	6.13	4.32	3.41

- c) (i) Without any calculations, what is the order of the above reaction? Give reasons.
(ii) Determine graphically the rate constant for the above reaction.
d) Find the half life for the reaction above.
e) At what time will the concentration of HCl be 7.22ml?
2) Two gases A and B react according to the equation: $A_{(g)} + 2B_{(g)} \rightarrow AB_{2(g)}$. A series of kinetics experiments:

Experiment	Initial concentration of A/mol dm ⁻³	Initial concentration of B/mol dm ⁻³	Initial rate/mol dm ⁻³ s ⁻¹
1	0.0500	0.100	1.00×10^{-1}
2	0.050	0.200	3.92×10^{-1}
3	0.100	0.100	1.95×10^{-1}

- a) Calculate, showing your working, the orders of reaction with respect to A and to B.
b) Write the rate equation for the reaction.
c) Calculate the rate constant, k, for the reaction using the results from experiment 3. State its units.
d) Explain, in terms of collision theory, why the rate of reaction increases with an increase in temperature.

DEPARTMENT OF CHEMISTRY
CHM 417: REACTION KINETICS

Instructions: Answer ALL Questions.

Time: 50 minutes

- a) Derive the rate equation and the half-life expression for the reaction $2A \rightarrow \text{products}$.
b) In the hydrolysis of ethyl acetate using equal concentrations of ester and NaOH, the following results were obtained:

T(minutes)	0	5	15	25	35
HCl (ml)	16.0	10.24	6.13	4.32	3.41

- c) (i) Without any calculations, what is the order of the above reaction? Give reasons.
(ii) Determine graphically the rate constant for the above reaction.
d) Find the half life for the reaction above.
e) At what time will the concentration of HCl be 7.22ml?
2) Two gases A and B react according to the equation: $A_{(g)} + 2B_{(g)} \rightarrow AB_{2(g)}$. A series of kinetics experiments:

Experiment	Initial concentration of A/mol dm ⁻³	Initial concentration of B/mol dm ⁻³	Initial rate/mol dm ⁻³ s ⁻¹
1	0.0500	0.100	1.00×10^{-1}
2	0.050	0.200	3.92×10^{-1}
3	0.100	0.100	1.95×10^{-1}

- a) Calculate, showing your working, the orders of reaction with respect to A and to B.
b) Write the rate equation for the reaction.
c) Calculate the rate constant, k, for the reaction using the results from experiment 3. State its units.
d) Explain, in terms of collision theory, why the rate of reaction increases with an increase in temperature.

Paper 4 of 4 — CHM 417 REVISION QUESTIONS

Level 400 • First Semester • PastQuestion

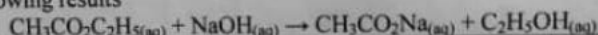
This source paper is a scanned document; its pages are reproduced below.

CHM 417 REVISION EXERCISES

- The decomposition of ethane (C_2H_6) to methyl radicals is a first-order reaction with a rate constant of $5.36 \times 10^{-4} s$ at $700^\circ C$. Calculate the half-life of the reaction in hours.
- The thermal decomposition of N_2O_5 obeys first-order kinetics. At $45^\circ C$, a plot of $\ln[N_2O_5]$ versus t gives a slope of $-6.18 \times 10^{-4} min^{-1}$. What is the half-life of the reaction?
- The decomposition of ethane (C_2H_6) to methyl radicals is a first-order reaction with a rate constant of $5.36 \times 10^{-4} s$ at $700^\circ C$. Calculate the half-life of the reaction in hours.
- a) A first-order reaction is 40% complete at the end of 5min. What is the reaction rate in seconds?
b) What is the molecularity of the following reactions?
i. $2NO + O_2 \rightarrow 2NO_2$
ii. $2FeCl_3 + SnCl_2 \rightarrow 2FeCl_2 + SnCl_4$
iii. $CH_3COOC_2H_5 + NaOH \rightarrow CH_3COONa + C_2H_5OH$
- A first order reaction is 38.5% complete in 480 seconds.
(i) Calculate the rate constant.
(ii) What is the value of the half-life?

How long will it take for the reaction to go to 75% completion?

- A study of the alkaline hydrolysis of ethyl ethanoate in aqueous solution shown in the reaction below gave the following results



Time/secs	0	224	377	621
Conc. of NaOH (mol/dm ³)	0.02578	0.02256	0.02101	0.01921
Conc. of ester (mol/dm ³)	0.01211	0.00889	0.00734	0.00554

Calculate the rate constant for this second order reaction.

- a) In the hydrolysis of ethyl acetate using equal concentrations of ester and NaOH, the following results were obtained:

T(minutes)	0	5	15	25	35
V _{HCl} (ml)	16.0	10.24	6.13	4.32	3.41

- (i) Without any calculations, what is the order of the above reaction? Give reasons.
(ii) Determine graphically the rate constant for the above reaction.
 - Find the half life for the reaction above.
 - At what time will the concentration of HCl be 7.22ml?
- The hydrolysis of ethyl nitrobenzoate by aqueous sodium hydroxide was followed at $25^\circ C$ by titration of the hydroxide at different stages of the reaction with $0.01 mol L^{-1} HCl$. If the initial concentration of both reactants is $0.01 mol L^{-1}$, show that the reaction is second order and calculate the rate constant

Time (s)	95	140	222	334	805
Titre (cm ³)	9.3	9.0	8.5	7.9	6.1

- The following kinetic data were obtained for the reaction between sodium thiosulphate and methyl iodide at $25^\circ C$, concentrations being expressed in arbitrary units.

Time (min)	0	4.75	10	20	35	55
$Na_2S_2O_3$	35.35	30.5	27.0	23.2	18.6	17.0
CH_3I	18.25	13.4	9.9	6.1	3.2	1.5

Determine the rate constant for the reaction.

- In the saponification of ethyl acetate in sodium hydroxide solution at $30^\circ C$ according to the equation: $CH_3COOC_2H_5 + NaOH \rightarrow CH_3COONa + C_2H_5OH$; the initial concentration of ester and alkali were both $0.5 mol/L$ and the decrease in ester concentration, x , was measured at the following times.

x (mmol/L)	5.91	11.42	16.3	22.07	27.17	31.47	36.44
Time (min)	4	9	15	24	37	53	83

- Find the rate constant for the reaction.
- Determine the half-life for the process